

Functional brief – iPad (IOS/Android) -based registration and document signing module

1. Implementation objective

The objective is to implement in the CM7med system a module that will allow:

- patient registration without paper forms,
- completion of patient data on an iPad,
- automatic generation of registration documents,
- signing documents on an iPad,
- saving data and signed documents into the patient's electronic medical documentation,
- future expansion to support signing of other documents in the facility, including treatment/procedure consent forms.

The module should be designed so that it is:

- simple for reception staff,
- simple for the patient,
- as cost-effective as possible to implement,
- easy to expand,
- secure and stable in the long term,
- suitable for storing documentation for many years and retrieving it later.

2. General assumption

The current system flow must remain unchanged.

Currently:

- a visit record (visit_id) is created first,
- if the patient is new, a patient_id is created after registration is completed,
- if the PESEL already exists in the database, the system allows the visit to be assigned to the existing patient and work on that patient's data.

This should remain unchanged.

The new module is intended to be an additional option within the existing flow, not a completely new registration mechanism.

3. Registration modes

After entering visit registration, the system should allow two modes:

3.1. Manual registration

The current mode, unchanged.

Reception staff manually enters patient data and completes registration as they do now.

3.2. Registration via iPad

A new mode.

Reception staff starts an iPad registration session for a specific visit.

4. Recommended iPad session model

Recommended simple solution:

- after selecting the “Registration via iPad” option, the system automatically generates a PIN / session token,
- the PIN is linked to a specific visit_id,
- this PIN is entered on the kiosk page on the iPad,
- after correct PIN entry, the system launches the form assigned to that visit.

This is the recommended proposal, but the developer may propose another equally simple and secure solution for linking the session to the visit.

5. PESEL logic and patient type detection

5.1. Basic rule

PESEL is the main field identifying the patient.

After entering the PESEL, the system should:

- validate the PESEL,
- determine the date of birth,
- determine the patient's age,
- determine whether the patient is an adult,
- determine whether the patient has reached the age of 16.

Based on this, the system should:

- show or hide the relevant fields,
- decide which set of documents to generate,
- decide which signatures are required.

5.2. If the PESEL does not exist in the database

The system proceeds with the full new-patient registration form on the iPad.

5.3. If the PESEL exists in the database

Recommended model:

- the system displays a simple message on the iPad, for example:
“The patient is already registered in the system. Please hand the device to reception.”

- the form does not continue independently,
- further handling is taken over by reception staff within the existing system flow.

That means:

- the logic of assigning the visit to an existing patient remains on the reception side,
- the iPad does not independently handle the full flow for an existing patient,
- on the reception side, patient data can be loaded, the visit can be assigned, and the iPad can optionally be handed back only for signing documents or completing missing data.

This is recommended as the simplest and safest solution.

6. iPad form logic

The form should be one dynamic form, not several separate forms.

Based on PESEL and age, the system should automatically switch logic between:

- adult patient,
- minor patient,
- minor patient aged 16+.

6.1. For an adult patient

The patient completes:

- first and last name,
- PESEL,
- date of birth (calculated or displayed by the system),
- residential address,
- phone number,
- email address.

Residential address should be split into separate fields, in line with how it currently exists in the patient edit section of the system:

- street,

- postal code,
- city,
- province / voivodeship.

These data should be saved into the same fields that already exist in the system.

Phone number should be entered and saved in the same format as currently used in the system, i.e. divided into:

- country code / dialing prefix,
- phone number.

The patient then selects the consent checkboxes.

At the end, the system automatically creates the basic set of documents for an adult patient.

6.2. For a minor patient

Once the system detects that the patient is a minor, it should display additional fields.

Important: in this scenario, the patient is the child, so the form must begin with the child's data, not the adult's data.

So first:

- child's data,
- then legal guardian / factual caregiver data,
- then the relevant consents and signatures.

These data should not only be used in documents, but also saved / updated in the CM7med system.

7. Mandatory documents

7.1. Adult patient – mandatory documents

1. Consent to the processing of personal data

Fields:

- patient's first and last name
- PESEL
- date of birth
- residential address:
 - street
 - postal code
 - city
 - province / voivodeship
- phone number:
 - country code / dialing prefix
 - phone number
- email address

Checkboxes:

- organization of healthcare services (including visit reminders)
- pro-health campaigns and initiatives
- newsletter / marketing information

Signature:

- adult patient's signature

2. Patient statement consenting to examination or other healthcare service

Fields:

- patient's first and last name
- PESEL

Signature:

- adult patient's signature
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7.2. Minor patient – mandatory documents

1. Statement of legal representative / factual caregiver

Fields:

- child's first and last name
- child's PESEL
- legal representative's / factual caregiver's first and last name
- legal representative's / factual caregiver's PESEL
- representation type:
 - legal representative
 - factual caregiver

Additionally, the system should provide the ability to define the person's relationship / role:

- mother
- father
- legal guardian
- other person (with the possibility to enter a custom value)

Additionally, the system should allow entering the data of **two parents / guardians**, if needed.

For each parent / guardian, the following fields should be available:

- first and last name
- PESEL
- phone number:
 - country code / dialing prefix
 - phone number
- relationship / role:
 - mother / father / legal guardian / other person

Signature:

- legal representative's / factual caregiver's signature

2. Consent to the processing of personal data – minor patient version

Fields:

- child's first and last name
- child's PESEL
- child's date of birth
- child's residential address:
 - street
 - postal code
 - city
 - province / voivodeship

Legal representative's / factual caregiver's data:

- guardian's first and last name
- guardian's PESEL
- phone number:
 - country code / dialing prefix
 - phone number
- guardian's email address
- basis of representation

Checkboxes:

- organization of healthcare services
- pro-health campaigns and initiatives
- newsletter / marketing information

Signatures:

- if the patient is under 16:
 - only the legal representative's / factual caregiver's signature
- if the patient is 16 or older:
 - separate patient signature
 - separate legal representative's / factual caregiver's signature

This must be handled conditionally and unambiguously.

Additional rule regarding the phone number:

- if the patient is under 16, their own phone number should not be saved as the main patient phone number; the phone number should instead be saved under the parent / guardian, and one of the parents / guardians should be marked as the main contact for the patient,
- if the patient is 16 or older, then their phone number may be shown and saved as the patient's main phone number, and additionally the phone numbers of the parents / guardians may also be saved.

8. Optional documents

Optional documents should not appear immediately as mandatory fields.

They should be handled as expandable sections or buttons such as:

- add authorization to obtain information about health status,
- add authorization to obtain medical documentation.

After selecting the relevant option, the system displays additional fields, and only then is the corresponding document added to the final package.

8.1. Patient statement authorizing a close person to receive information about health status and services provided

Fields:

- patient's first and last name
- patient's PESEL
- close person's first and last name
- close person's PESEL
- close person's address
- close person's phone number

Signature:

- patient's signature

8.2. Patient statement authorizing a close person to obtain medical documentation

Fields:

- patient's first and last name
- patient's PESEL
- close person's first and last name

- close person's PESEL

Signature:

- patient's signature
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9. Document generation rule

The system should work as follows:

- after completing the data and required checkboxes, the system always automatically generates the basic set of documents for the relevant patient type,
- optional documents are added only when the user selects the relevant section and completes the required fields.

In other words:

- basic package — always,
- extended package — only conditionally.

This should be as simple as possible for both the patient and reception staff.

10. Signatures

10.1. General rule

The patient signs at the end of the process.

10.2. Recommended model

Initially, the recommended model is:

- signatures at the end of the process,
- no mandatory PDF preview before signature,
- final documents generated after signature.

10.3. Signature of a 16+ patient

If the patient is at least 16 years old and still a minor, the system must require:

- a separate patient signature,
- a separate legal representative's / factual caregiver's signature.

This must be clearly separated in the form and in the documents.

10.4. Legal and secure signature handling

The signature must be stored in such a way that it is protected, integral, attributable to the specific document version and signing event, and treated legally as the patient's or guardian's signature within the system process — not merely as a loose image or standalone photo file.

This means the developer should implement signature storage in a form that:

- is bound to the specific document and document version,
- is bound to the patient / guardian and visit context,
- is timestamped,
- cannot be freely swapped or replaced outside the document lifecycle,
- is preserved as part of the legal record of the signed document.

The signature should therefore be treated as a protected signed-record element of the document workflow, not as an ordinary uploaded picture.

11. Final button

The final button in the process should be:

- “Complete registration”

or, in the office / procedure mode:

- “Sign and complete”

After clicking this button, the system should automatically:

- save / update patient data,
 - save / update guardian data where applicable,
 - save checkboxes and consents,
 - generate the complete set of documents,
 - save the signature,
 - generate the final PDF / PDFs,
 - assign the documents to the patient and visit,
 - end the iPad session.
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12. Storage format and documents

12.1. Database data

The source of truth should be structured data saved in the database.

The database should store:

- patient data,
- guardian data,
- authorized person data,
- consent checkbox values,
- document status,
- template version,
- signature,
- linkage to patient and visit.

12.2. Final PDF

After signing, the system should generate the final PDF.

Ultimately, the PDF should automatically be placed in the patient’s / visit’s documents section.

If, at the MVP stage, integration with the current documents section turns out to be too complex, a simplified variant is acceptable:

- the system generates the PDF,
- reception / staff has a button:

“Download patient documents”

- the document may be downloaded as the final legal document.

12.3. HL7 / other standards

At this stage, we do not require these documents to be stored as HL7.

Recommended model:

- structured data in the database,
- final PDF,
- possibility of later adding export to interoperability standards.

For future discussion remains the question whether documents or parts of them should also later be stored and/or exported in interoperable formats (e.g. HL7 / CDA / FHIR).

13. Versioning and corrections

13.1. Before registration is completed

The user must be able to go back through the form and correct fields.

13.2. After registration is completed

A signed document may not be edited directly.

If something needs to be corrected after registration is completed, the system should provide an action such as:

- “Correct data and generate again”

or

- “Complete document again”

Then:

- the system creates a new document version,
- the old version remains in history,
- the old version is not deleted,
- patient data is updated,
- the new version requires a new signature.

This should be as simple and safe as possible.

14. Information that must be saved for each document version

For each document version, the system must save:

- document identifier,
- version identifier,
- document type,
- template version,
- patient identifier,
- visit identifier,
- creation date,
- creation time,
- signature date,
- signature time,
- who initiated the process,
- from which device the signature was made,
- document status,
- link to the previous version if it is a correction.

15. Document statuses

Recommended statuses:

- draft,
 - in progress,
 - ready for signature,
 - signed,
 - replaced by a new version,
 - cancelled.
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16. Long-term storage

The system should be designed so that it:

- stores documentation for many years,
- allows data and documents to be recovered at any time,
- preserves integrity and version history,
- does not rely solely on a single PDF,
- has both structured data saved in the database and the final document in PDF form.

This is intended to be a durable and secure solution for years.

17. Further step for later clarification

At the current stage, not all future cases are being finalized.

To be determined at the next stage:

- exact screen-by-screen view,
 - exact statuses on the reception and doctor side,
 - exact structure of the documents tab,
 - full handling of additional document types,
 - possible export integration.
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18. Additional point for future implementation – documents in the doctor's office

The system should be designed so that, on similar principles, a doctor may later use their own iPad in the office to sign documents such as:

- procedure consent,
- examination consent,
- other medical documents generated by the system.

In this scenario:

- the patient is already registered,
- the data is already in the system,
- the document is generated automatically from patient, visit, service, and doctor data,
- the patient only reviews the content and signs on the iPad,
- the document is saved analogously to registration documents.

This should be foreseen architecturally now, even if it will not be implemented in the first stage.

19. Security and access to the form

Strong process security must be provided so that the form is not accessible in an open and unlinked manner.

There must not be a situation in which any person can simply open the form page on an iPad, Android tablet, or another device and gain access to the registration process without an authorized session.

Recommended security assumptions:

- the form must be linked to an active session started by reception staff or a doctor,
- access to the form must take place through a PIN / token / other method linked to a specific visit_id,
- the session should be time-limited,
- after the process is completed, the session should be closed,
- after completion or interruption, the screen should return to the start / kiosk screen,
- the form must not allow viewing data of other patients,
- the system should restrict the possibility of re-entering the same session after it has been closed,
- appropriate security mechanisms for kiosk-device operation should be provided.

The exact technical implementation of these security measures remains for the developer to propose, but the assumption is that the solution must be secure, linked to the reception / doctor session, and resistant to unauthorized access.

